7 Autism

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7.1 Notes on Language

The last decade has seen autism at the center of a significant paradigm shift, shaping our understanding and approaches to supporting individuals diagnosed with autism. Our language reflects, but also shapes, these ongoing paradigm shifts. In this chapter, we believe providing a note and explanation for the language used throughout is necessary, first, regarding our use of "identity-first" language, and second, in relation to neurorehabilitation.

Person-first language (i.e., a person with autism) is often how we refer to clinical populations, separating the individual from their diagnosis [1]. However, many members of the autistic community currently reject the notion that autism can be viewed as separate from their identity and instead express a preference for identity-first language (i.e., autistic person). Identity-first language is proposed to

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acknowledge that autism is integral to an individual's identity and should be tolerated and valued. Although there is no consensus on the matter, several empirical studies suggest that identity-first language may be the preferred terminology of many English-speaking autistic individuals [2, 3]. Aligning with these preferences, we employ identity-first language in this chapter. However, it is important to note that most research on this topic has been conducted primarily in English-speaking Western countries and therefore may not represent the perspectives of individuals globally. The autistic population is also not monolithic; perspectives and language preferences will vary from individual to individual. For this reason, it is considered best practice to enquire and use the language preferences of the individuals themselves.

The term "neurorehabilitation" in the context of autism also poses a need for discussion. Dividing the term into its constituent components produces-"neuro" referring to the function of the central nervous system, "habilitation" referring to the development and optimization of skills and abilities, and "re" signifying relearning or redeveloping those skills or abilities that had been previously acquired. Broadly defined, neurorehabilitation thus denotes a series of approaches or processes that aim to compensate, minimize, or optimize outcomes in the face of functional challenges associated with nervous system injury or impairment [4]. Autism is a neurodevelopmental condition characterized by altered neurological structure and function; thus, neurorehabilitation seems relevant from this perspective. On the other hand, "rehabilitation" implies that some skill or ability must be relearned, recovered, or restored. More contemporary perspectives, however, view autism as a functional difference rather than a loss. Thus, there is nothing to restore or recover per se. Instead, we focus on habilitation, which is concerned with developing, improving, and optimizing functioning and skills. To acknowledge these considerations, we use the term "neuro(re)habilitation" to emphasize that the support approaches we apply seek to habilitate, not rehabilitate.

7.2 Autism

Autism is today defined by the two diagnostic manuals: the fifth edition of the Diagnostic and Statistical Manual for Mental Disorders (DSM-5) [5] and the 11th edition of the International Classification of Diseases (ICD-11) [6]. Autism or "Autism Spectrum Disorder" falls under the umbrella of neurodevelopmental conditions, a broad group inclusive of other conditions such as Attention Deficit Hyperactivity Disorder (ADHD), Intellectual Disability, Communication Disorders, Neurodevelopmental Motor Disorders, and Specific Learning Disorders.

Much like other neurodevelopmental conditions, the precise causes of autism remain a topic of investigation, but they are likely multifactorial, including genetic, neurological, and environmental factors and their interactions. Autism is known to be a highly heritable condition, meaning that there is a strong genetic influence underlying its development [7]. Environmental conditions, such as prenatal and perinatal factors, are likely to also play a role in the development of autism through complex gene-environment interactions [8]. Autism is one of the most common neurodevelopmental conditions in the present day. Global prevalence rates currently estimate that one in 100 children have an autism diagnosis [9]. These prevalence rates represent a dramatic increase in diagnosis, with autism previously thought to be a relatively rare condition. The reason underlying increasing diagnostic rates is likely manifold, probably driven foremost by increasing public understanding and knowledge, and a broadening of diagnostic criteria [9]. Regardless of the reasons underpinning increasing diagnostic rates, these figures indicate a significant need for intervention and support approaches that reduce disability and promote positive functioning and health outcomes.

Autism is not defined by any specific biomarker, and thus, diagnosis is based on behavioral observation and the presence of specific characteristics. These characteristics are believed to be developmental in nature and present from birth, but are often not immediately observable. Autism can be diagnosed in children as young as 2 years of age. However, many individuals are not diagnosed until later childhood, adolescence, or even adulthood, when environmental demands begin to exceed the individuals' capacities.

Two core diagnostic criteria must be met for an autism diagnosis: (1) persistent difficulties in social communication and interaction, and (2) restricted or repetitive behaviors and interactions. The first criterion requires persistent difficulties with social communication and interaction across multiple contexts. These difficulties can manifest in many ways, but must be observed in three categories: (1) social-emotional reciprocity, (2) non-verbal communication, and (3) developing, maintaining, and understanding relationships. The second criterion requires the presence of restricted or repetitive patterns of behavior, interest, or activity. The specific manifestation of this domain can vary, but to receive a diagnosis, an individual must show behaviors that meet at least two of the four categories: (1) Stereotyped or repetitive movements, use of objects, or speech, (2) insistence on sameness, inflexible adherence to routines, or ritualized patterns of verbal or nonverbal behavior, (3) restricted, fixated interests, and (4) hyper- or hypo reactivity to sensory input or unusual sensory interests. These characteristics must be present in early development (although they may not be evident until later in life) and cause clinically significant impairment in functioning across social, occupational, or other domains.

7.3 Key Concepts and Considerations

Understanding autism and approaches to neuro(re)habilitation and support involves several key concepts and considerations, including heterogeneity, the role of sex and gender, and the impact of co-occurring conditions.

7.3.1 Heterogeneity: A Spectrum Condition

Autism, though defined by common core diagnostic criteria, is highly variable and heterogeneous. Heterogeneity can be observed between individuals regarding their underlying neurobiological mechanisms, phenotypic presentations, and functional impacts, as well as within individuals, which evolves across the lifespan and dynamically across different contexts. A key concept required for understanding autism is thus heterogeneity.

The between-individual heterogeneity of autism has been reflected differently in various iterations of the diagnostic manuals. In the DSM-IV and ICD-10, autism was not considered a singular condition, but rather a collection of conditions grouped under the category of "Pervasive Developmental Disorder" [10]. These diagnostic categories sought to capture heterogeneity by describing the degree to which the different diagnostic criteria were met and the extent of their functional impacts. However, this led to often inconsistently applied diagnoses, and the distinction between different diagnoses was not well supported by evidence [11]. The most recent versions of the DSM and ICD (DSM-5 and ICD-11) now subsume previous diagnoses under the unified diagnosis of "Autism Spectrum Disorder[1] [5, 6], capturing heterogeneity by introducing severity levels and specifiers. Severity is divided into three levels based on an individual's support needs, ranging from level 1: "Requiring support," level 2: "Requiring substantial support," and level 3: "Requiring very substantial support." The DSM-5 also includes other specifics, such as whether it is accompanied by intellectual disability or language impairment.

Although diagnostic manuals typically account for heterogeneity through this unidimensional spectrum of support needs, it should not be misinterpreted that the difficulties and abilities experienced by autistic individuals are uniform across domains. Rather, some evidence suggests that the two core diagnostic domains may be dissociable at the genetic [12], neural [13], and phenotypic [14] level, and autism has been associated with several comparative strengths [14]. Thus, the manifestation of difficulties and abilities often appears as a "spiky" or "uneven" profile. This complex profile can be represented by a kaleidoscope, demonstrating its multidimensional nature. Autism can, therefore, be captured on two spectrums: first, a unidimensional spectrum of support needs ranging from requiring support to substantial support, and second, a multidimensional spectrum capturing individual strength and difficulty profiles.

Strength and difficulty profiles can also vary within individuals across the developmental trajectory, as different strengths and challenges may emerge over time due to neurological maturation, changing contexts and demands, and the development of skills and abilities. The phenomenon of variability across the lifespan has been referred to as chronogeniety [15]. Longitudinal evidence suggests that there can be different phenotypic trajectories over development. For instance, subgroups may

[1] With the exception of Rett's Syndrome, which is now considered a distinct neurodevelopmental condition following the identification of specific gene mutations underpinning its development.

have increasing social difficulties over time, while others may have decreasing social difficulties, with these trajectories possibly influenced by factors such as gender [16].

Key transition periods across the lifespan might contribute to the emergence of different strength and difficulty profiles, owing to new challenges, contexts, and opportunities. The first major transition period, commencing school, introduces new social, behavioral, and cognitive expectations. Children face increased demands for independence and must adhere to more formalized routines and structures of the learning environment while navigating new relationships with peers and educators. These social, behavioral, and cognitive demands continue to increase in complexity, particularly over the transition to adolescence and adulthood. Adolescence is a particularly critical juncture where the increased complexity of demands coincides with pubertal onset and significant neural remodeling. Hormonal influences, alongside changes in neural structure and function, affect a wide range of cognitive and emotional functions, with adolescence being a particularly vulnerable time for the development of mental health conditions [17].

The presentations and impacts of autism can also vary across different life contexts and situations. This variability in presentation is noted by the most recent version of the DSM-5, which accounts for the context-dependent nature of several observable autistic-like traits. For instance, an individual may have difficulties integrating verbal and non-verbal interaction (such as eye contact, speech, and gestures) only when required to do so for sustained periods or when under stress [5]. More recent work also highlights the influence of social contexts on the manifestation of difficulties. For instance, the "double-empathy problem" proposes that social difficulties are not due solely to impairments within an autistic individual but due to issues arising in the reciprocal interaction between autistic and non-autistic individuals [18]. Some evidence, although preliminary, indeed suggests that autistic-to-autistic communication may be similarly effective as non-autistic-to-non-autistic communication, with breakdowns occurring only when neurotypes (autistic and non-autistic) are mixed. Thus, there is some evidence that the communication partner influences the challenges experienced in social interaction [19].

Additionally, different strength and difficulty profiles can emerge as a result of various contexts, which is supported by a wide array of research that shows environments can facilitate or hinder meaningful participation for autistic individuals across multiple life domains, including employment, education, and community participation. Evidence consistently points to the role of social attitudes and supports, the accessibility of accommodations, and other physical and societal environmental factors in influencing an individual's functioning [20, 21].

As phenotypic presentations and functional profiles can vary considerably between and within individuals across the lifespan and various contexts, one-size-fits-all neuro(re)habilitation approaches are unlikely to be effective, and more individualized approaches are required. Neuro(re)habilitation approaches must be adaptive and flexible to individual needs and contexts, and should consider how different strengths, abilities, and challenges may emerge or diminish over time.

7.3.2 Sex and Gender: Influences on Profiles and Needs

Autism was, until relatively recently, believed to be a predominantly male condition with a gender ratio of 4:1. However, recent evidence suggests that this gap may be much smaller than previously estimated [22, 23]. There is a growing recognition that many girls and women have historically gone undiagnosed or misdiagnosed due to male-oriented descriptions of the condition and other neurobiological, social, and cultural influences [24]. Sex and gender differences have been observed in the manifestation of autism characteristics and the associated experiences and challenges of autism. For example, some evidence has suggested that girls and women may have higher social motivation and fewer social difficulties, but still experience difficulties maintaining long-term relationships and managing relationship conflicts [23]. Other evidence, although mostly preliminary, suggests that girls and women may be more likely to engage in camouflaging behaviors than boys and men, potentially masking characteristics, delaying detection, and leading to more long-term mental health outcomes [25]. As sex and gender can influence the profile of autism and can result in different needs, neuro(re)habilitation approaches must be sensitive to these differing needs.

7.3.3 Co-occurring Conditions: Compounding and Dual Impacts

Autism commonly co-occurs with other neurodevelopmental, mental health, and physical health conditions, which can have compounding and negative influences on functioning. Attention Deficit Hyperactivity Disorder (ADHD) is one of the most frequently co-occurring conditions seen with autism, with some estimates suggesting that the co-occurrence may be as high as 50–70% [26]. Co-occurring autism and ADHD are generally associated with greater impairment [27] and can influence the effectiveness of intervention and support. For example, recent evidence has suggested that social skill group training may be less effective for autistic individuals with co-occurring ADHD [28]. At the same time, autistic individuals are also at a higher risk of experiencing mental health challenges and have a high prevalence of psychiatric conditions such as anxiety and depression [29], and are significantly more likely to have other co-occurring neurological disorders such as epilepsy, migraine/headache, cerebral palsy, macrocephaly, hydrocephalus, and other congenital abnormalities of the nervous system than the general population [30]. As co-occurring conditions are commonly observed in autistic populations, can compound functional impairment, and influence the efficacy of intervention and support; neuro(re)habilitation must consider the presence and impact of co-occurring conditions.

7.3.4 Competing and Complementary Approaches: Models of Disability

Disability in the context of autism can be conceptualized through different, and often thought to be competing, models. Through the lens of these models, our view of disability in the context of autism and the causative reasons for the challenges experienced by autistic individuals is shaped, influencing neuro(re)habilitation approaches.

Biomedical models of disability are perhaps the most well-established and prominent models guiding our understanding of autism to date. Biomedical models view disability as arising primarily from physiological or biological malfunction within an individual. Although broader social and psychological factors are incorporated into more contemporary biomedical models, they still predominantly focus on biological or physiological mechanisms causing impairment. Neuro(re)habilitation rooted in biomedical models tends to focus on implementing interventions at an individual level, such as through the provision of medication.

The social model of disability, on the other hand, challenges the more medicalized perspective of disability. Where biomedical models propose that disability originates from impairments within an individual, social models emphasize the role of environments, including the physical, social, attitudinal, and political environment, which can influence functioning and disability. Through a social model lens, neuro(re)habilitation approaches focus on environmental changes such as accommodations and advocate for greater understanding and acceptance.

Biomedical and social models of disability offer their own advantages concerning conceptualizing disability. Biomedical models are necessary for diagnosis and for understanding and treating various psychiatric and physical health conditions, while social models are necessary for understanding the role of environmental factors in shaping disability and for developing accommodations and other environmental interventions and supports. When used independently, however, they also carry limitations; primarily, neither approach fully accounts for the range of factors influencing disability, the interactions that occur between an individual and their environments, and the diversity and variability inherent to the human experience.

Unlike biomedical and social models of disability, which place more specific emphasis on either biological or environmental causes of disability, bio-psycho-social models place equal weight on both, viewing health and disability as arising from the interaction between multiple systems: biological (physiological), psychological, and environmental factors. A common example of a bio-psycho-social approach to disability is the International Classification of Functioning, Disability and Health (ICF), which effectively reconciles both biomedical and social models of disability [31, 32].

The models through which disability in autism is viewed can fundamentally alter approaches to neuro(re)habilitation. Approaches rooted in more biomedical perspectives may focus on individually oriented approaches that seek to remediate deficits and enhance individual skills and abilities. On the other hand, those rooted in more social perspectives will focus on environmental adaptation. We advocate for bio-psycho-social approaches to understanding functioning and disability in autism, providing a holistic view of not only individual challenges, but also strengths, and environmental barriers and facilitators.

7.4 Neuro(re)habilitation in the Context of Autism

Given the many considerations of neuro(re)habilitation in autism, including its heterogeneous profile within and between individuals, different conceptualizations of difficulties, and impacts of sex, gender, and co-occurrence, providing an in-depth profile of the range of neuro(re)habilitation approaches within the field of autism is not possible within the scope of this chapter. Instead, we discuss key principles and practices underpinning neuro(re)habilitation approaches in autism.

7.4.1 Neuro(re)habilitation Targets

Intervention and support targets for neuro(re)habilitation can vary widely and are influenced by the model of disability through which autism is viewed. Still, ultimately, their goal is to improve an individual's quality of life and functioning. Neuro(re)habilitation targets can focus on individual factors (those related to the psychological or cognitive functioning of the individual), their activities or participation, or their environments. Here, we organize common neuro(re)habilitation targets in autism using the bio-psycho-social framework of the International Classification of Functioning, Health and Disability (ICF) [31] (Fig. 7.1).

The topic of neuro(re)habilitation targets has been subject to debate and is central to ongoing paradigm shifts in the field, characterized by a lessening focus on biomedical models and an increasing popularity of the social and bio-psycho-social models of disability. Where previously there was a significant focus on reducing autistic behaviors (e.g., repetitive behaviors) alone, there is growing recognition that intervention and support should place equal emphasis on more meaningful outcomes (such as participation) and environmental adaptation and modification.

7.4.2 Principles and Practice

7.4.2.1 Neural Plasticity and Windows of Opportunity

As autism is a condition arising from atypical neurological functioning and maturation, with evidence indicating atypicalities in the functional and structural organization of the brain, neural plasticity is central to neuro(re)habilitation. Neuroplasticity refers to the ability of the central nervous system to mature and adapt dynamically across the lifespan in response to environmental factors or injury. It is an essential function of the brain, providing the necessary foundation for learning, memory, adaptation, and recovery from injury. Neural plasticity is driven by a range of mechanisms that operate at various levels of neural architecture and function. In short,

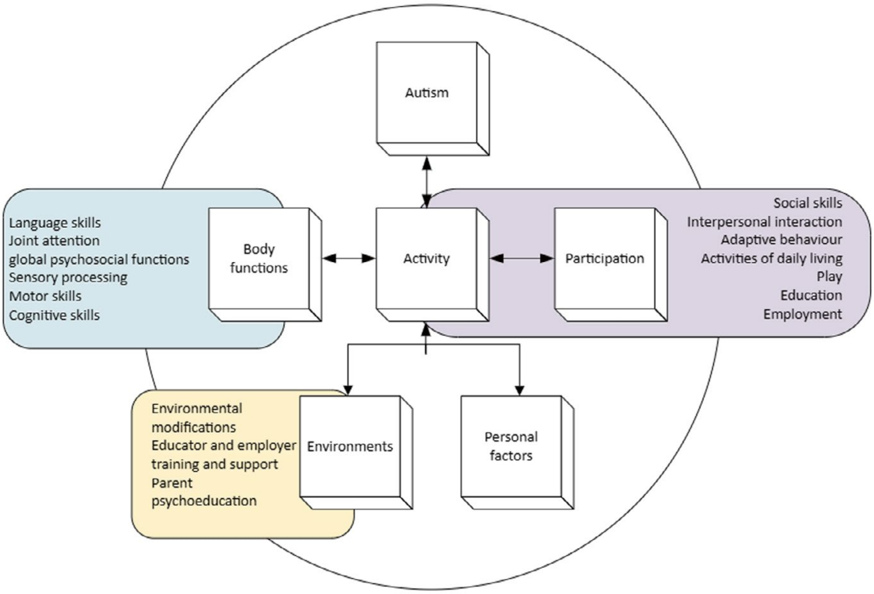

Fig. 7.1 Common neuro(re)habilitation targets organized according to the WHO ICF

repeated exposures to internal and external stimuli, such as exposure to language as a child, activate neural circuits. With repeated activation, these circuits strengthen neural connections, thereby enhancing the effectiveness of information transfer. On the other hand, when neurons are less frequently activated due to reduced exposure to external stimulation, the synaptic connections become weaker over time (long-term synaptic depression) [33, 34].

Although the brain remains plastic and able to learn and adapt throughout the lifespan, there are specific time-sensitive developmental periods where the brain is especially plastic and malleable, presenting "windows of opportunity" for neuro(re)habilitation [34, 35]. Early childhood, particularly the first 3–5 years, represents the most significant window of opportunity, where the brain is highly receptive to learning experiences and environmental exposures, both internal and external [36]. For this reason, there is often a significant focus on early intervention during these years. Adolescence represents the second significant window of opportunity. Although during this period there is a greater shift toward stabilizing existing neural circuits, plasticity becomes heightened again, as pubertal changes trigger neural remodeling, involving neural reorganization, synaptic pruning, and myelination. The increase in neural plasticity during adolescence makes adolescents particularly responsive to environmental exposures, both those that are adverse and those that are supportive. Adolescence thus provides an opportunity to provide experiential learning experiences that can shape future development [37].

The brain's ability to continuously and dynamically learn and adapt across the lifespan presents opportunities for neuro(re)habilitation across all life stages.

Understanding neural plasticity, the role of learning opportunities and environmental exposures, and particularly sensitive windows of time, can inform the timing and focus of neuro(re)habilitation approaches.

7.4.2.2 Intensity and Duration

Neuro(re)habilitation intervention can vary regarding duration (length of intervention period) and intensity (amount of intervention over a given period). The consideration of intervention intensity and duration is essential when implementing neuro(re)habilitation approaches.

Generally, the extent of intervention delivered seems to be associated with intervention outcomes, and there is a significant focus on providing particularly intensive intervention in the early childhood years. This stems from the belief that such intensive intervention during the early critical years of neural plasticity will lead to greater long-term outcomes. A common example of an intervention delivered during early childhood is Early Intensive Behavioral Intervention (EIBI), which is typically delivered for 20–40 h per week over multiple years [38]. Most evidence for the positive impact of intervention intensity and duration on outcomes stems from studies of behavioral approaches [36], but has also been found in other intervention areas, such as social skills training programs, where longer variants perform better than shorter variants [39]. On the other hand, however, a recent meta-analysis of early childhood autism interventions across a range of intervention approaches and outcome types did not find any robust evidence to support the notion that intervention amount influences outcome [40]. These findings suggest that there is a need to consider the nuances of intervention intensity and duration rather than assuming a greater extent of intervention will inherently lead to improved outcomes.

There is also a need to consider the intensity and duration of intervention delivery within the context of an individual's life situation. For instance, while potentially beneficial, intensive intervention over extended time periods can be costly and demanding for the individual and their family, and may detract from the individual's participation in key daily life activities, particularly for school-aged children. Other research exploring the efficacy of Cognitive Behavioral Therapy (CBT) found that intervention engagement was a key predictor of outcome [41], suggesting that other factors beyond intensity and duration are important to consider when delivering intervention. Evidence thus seems to indicate a positive influence of intervention and duration on outcome; however, other factors, such as engagement, should also be considered. Additionally, decisions regarding intervention intensity and duration should be weighed alongside consideration of financial and logistical demands, as well as impacts on daily life participation for the individual and their families.

7.4.2.3 Generalizability

Core to neuro(re)habilitation and closely linked to neural plasticity is generalization. Generalization is the ability to transfer skills and abilities learned in the neuro(re)habilitation context to real-life situations. Generalization is thus critical to consider for any neuro(re)habilitation approach to have a meaningful and beneficial impact on the individual in their day-to-day life. Some behavioral interventions, such as Discrete Trial Training, based on the principles of Applied Behavioral Analysis (ABA), have been suggested to be less effective in promoting the generalization of skills to other contexts due to their highly structured nature. However, they may be particularly effective in supporting the development of targeted skills. More naturalistic interventions delivered in real-world environments may, on the other hand, be more effective in helping individuals transfer skills to other life domains. Often in neuro(re)habilitation, a balance must be made between opportunities for repetition, while also providing more real-life contexts through which to practice skills. An example of an approach to enhancing generalizability is the inclusion of "homework" to be done outside of clinical contact, providing individuals with opportunities to practice newly learned skills within real-world settings.

7.4.2.4 Contexts

Neuro(re)habilitation in autism is rarely undertaken within a singular, isolated context. Instead, it occurs across a range of contexts and involves a variety of individuals, reflecting the everyday life situations and needs of the individual. Interventions can be delivered to individuals, families, and broader communities and networks. Individually orientated interventions focus on targeting specific behaviors and focusing on skill development. A common example of interventions in this domain might be social skills group training programs, which are often delivered to groups of autistic individuals to support their social interaction and communication skill development. These programs are typically delivered in clinical contexts, but recent work has expanded these programs to more naturalistic environments, such as schools [42].

Interventions and support for autistic individuals can also be delivered to those surrounding the autistic individual, rather than to the individual themselves. Family- and parent-focused interventions, such as psychoeducation, are commonly implemented during the early childhood period. As individuals age, support often shifts toward modifications, adaptations, and training within educational and employment settings.

7.4.2.5 Key Neuro(re)habilitation Approaches Across the Lifespan

We have established that autism is a heterogeneous condition characterized by significant within- and between-person heterogeneity, which necessarily shapes neuro(re)habilitation approaches across the lifespan and provide an overview of key principles in neuro(re)habilitation. We now summarize key approaches and considerations across the lifespan (Fig. 7.2).

Early Childhood

Given that early development presents one of the strongest windows of opportunity for development, significant emphasis is placed on providing early evidence-based interventions and supports in the early childhood period. Indeed, a large body of evidence has shown that earlier intervention is associated with better long-term outcomes for autistic youth [43]. Early intervention aims to intervene while neural pathways are still forming, capitalizing on the brain's significant plasticity by

	Early Childhood	Childhood	Adolescence	Adulthood

	Early Childhood	Childhood	Adolescence	Adulthood
Focus	**Body Function** Speech and language development Co-occurring conditions Cognitive skills Sensory processing Fine and gross motor skills **Activity and Participation** Basic social interaction skills Basic self-care skills (e.g., toileting) Self-regulation skills Parent-child bonding **Environment** Parent psychoeducation Early learning support	**Body Function** Executive functioning skills Fine and gross motor skills Sensory processing Emotion regulation Co-occurring conditions **Activity and Participation** Peer interaction and relationship skills Self-regulation skills Developing independence in self-care (e.g., daily routine) **Environment** Inclusion in school and community Academic support	**Body Function** Executive functioning skills Fine and gross motor skills Sensory processing Emotion regulation Co-occurring conditions **Activity and Participation** Advanced interaction and relationship skills Self-regulation (coping strategies, complex emotions, handling responsibilities) Transition planning (vocational) Independence in daily living and self-care skills **Environment** Inclusion in school and community Academic and vocational support	**Body Function** Co-occurring conditions **Activity and Participation** Advanced interaction and relationship skills (colleagues, romantic relationships) Managing daily living activities (finances, household maintenance, self-care Independent living skills **Environment** Employment and study support Inclusion in community
Key Approaches	**Individual** Behavioural approaches (e.g., ABA, DTT, EIBI) Developmental approaches (e.g., DIR FloorTime, NDBI (e.g., ESDM) occupational therapy speech and language therapy Treatment and management of co-occurring conditions **Family** Family psychoeducation Parent-mediated interventions (JASPER) **Community** Preschool support Environmental modifications	**Individual** Behavioural approaches Social skill support (e.g., KONTAKT) CBT (e.g., Face your Fears) occupational therapy speech and language therapy (e.g., AAC) Treatment and management of co-occurring conditions **Family** Behavioural management support Supports in groups (such as social skills) **Community** Teacher training/education Education intervention Academic support/IEPs Environmental modifications	**Individual** Behavioural approaches Social skill support (e.g., KONTAKT) CBT (e.g., FYF) Specific disciplines occupational therapy speech and language therapy Treatment and management of co-occurring conditions **Family** Behavioural management support Supports in groups (such as social skills; e.g., KONTAKT) **Community** Teacher training/education Education intervention Academic support/IEPs Environmental modifications	**Individual** Social skill support (e.g., PEERS) CBT occupational therapy Treatment and management of co-occurring conditions **Family** Informal support **Community** Employer training/coaching (IEST) Mentorship programs (SPM) Environmental modifications

Fig. 7.2 Common foci of neuro(re)habilitation across the lifespan and key approaches across contexts. Inspired by The Lancet Commission on the future of care and clinical research in autism [43]. Figures show change in primary neuro(re)habilitation targets across the lifespan (Body functions = blue, Activity and participation = purple, and Environmental = yellow). Generally, there is a greater emphasis on body function in early childhood, with an increasing emphasis on activity and participation and environmental targets as individuals age. *ABA* Applied Behavioral Analysis [49], *DTT* Discrete Trial Training [50], *EIBI* Early Intensive Behavioral Intervention [38], *DIR FloorTime* Developmental, Individual-differences, and Relationship-based model FloorTime [46], *NDBI* Naturalistic Developmental Behavioral Interventions [51], *ESDM* Early Start Denver Model [52], *JASPER* Joint Attention, Symbolic Play, Engagement, and Regulation (JASPER) Intervention [48], KONTAKT [53], *CBT* Cognitive Behavioral Therapy, *FYF* Facing Your Fears: Group Therapy for Managing Anxiety in Children with High-Functioning Autism Spectrum Disorders [54], *AAC* Augmentative and Alternative Communication, PEERS [55], *IEST* Integrated Employment Success Tool [56], *SPM* Specialist Peer Mentoring [57]

providing early experiential learning opportunities. Often, approaches employed during this period seek to support the acquisition of skills and knowledge that align with the typical development trajectory. They focus on developing skills and knowledge that are believed to be important for development and that are identified as being delayed or absent in the individual child [44, 45]. Commonly targeted skills include language, motor, communication, and daily living skills. Some examples of common child-centered interventions implemented during this period include the Developmental, Individual-differences, and Relationship-based model (DIR Floortime) [46], Positive Behaviour Support (PBS) [47], and Early Intensive Behavioral Intervention (EIBI) [38]. Interventions in the early childhood period also place a particular emphasis on educating families through psychoeducation by providing families with information about autism and how to support their child's development. Parent-mediated interventions, such as JASPER [48], are also commonly implemented to coach and support parents in applying various intervention and support strategies within day-to-day life.

Childhood

In childhood, neuro(re)habilitation approaches are still implemented to support children in further developing the skills necessary for participation in major life areas. While interventions were delivered primarily to and through family and caregivers in early childhood, there is less emphasis on family-mediated approaches in childhood, as children move out of the home and into other contexts, such as school. Child-centered approaches include social skills training programs that seek to support youth in developing social interaction and communication skills. Additional personalized interventions may address other critical areas, such as emotion regulation, executive functioning, and daily living skills, with the goal of supporting children in navigating necessary academic, social, and community settings more independently. In the educational context, several adaptations and modifications can be made to promote success at school, such as educator training, academic support plans (such as Individualized Education Plans), and environmental modifications to reduce sensory demands and promote participation.

Adolescence

Adolescence represents a key transition period characterized by increased demands for autonomy and independence. As youth enter adolescence, there is an increased focus on supporting independence and autonomy as youth begin to age into adulthood. Neuro(re)habilitation during this period may include training in daily life skills, social skills programs with an emphasis on peer, romantic, and professional relationships, as well as transition planning. Mental health difficulties, such as anxiety, can become increasingly prevalent during this age period, and thus, specific intervention, such as Cognitive Behavioral Therapy (CBT), may become particularly helpful.

Adulthood

In adulthood, there is a focus on areas necessary for independent living and engagement in everyday life domains such as employment. There are also ongoing efforts to address co-occurring conditions, such as mental health issues. In adulthood, there is a significant shift in focus from the body function domain to promoting inclusion and participation in major life areas, as well as environmental modifications. Neuro(re)habilitation in this period might include further support in social skill development, as well as independent living skills, such as managing finances and household management. For some adults, this may also include exploration of supported living arrangements. This may include a specific focus on areas such as job coaching, workplace accommodations, and employer training.

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